

Exploratory Data Analysis – Practical Assessment Report

Assessment Tool 3 | Course: Fundamentals of Data Science (DSA04) | CO1: Role of Data Science & Big Data Ecosystem

Prepared by: **Shaik Muskan**

This report presents an Exploratory Data Analysis (EDA) of five real-world business datasets spanning education, healthcare, retail, banking, and agriculture. For each dataset the analysis covers structure and data types, missing-value and duplicate checks, descriptive statistics, visual exploration through bar charts, box plots, scatter plots, histograms and a correlation heatmap, and a set of data-driven observations and conclusions. The goal throughout is to uncover the factors that most strongly influence each outcome variable and to translate those patterns into insights that support practical decision-making.

1. Education Analytics Dataset

Purpose & Significance: This dataset lets a college examine how attendance, study habits and internal assessment performance relate to a student's final result. The insights support early identification of at-risk students and evidence-based academic interventions aimed at improving overall pass rates.

Column	Type	Description
Student_ID	Categorical	Unique identifier for each student
Attendance	Numerical	Class attendance percentage
Study_Hours	Numerical	Average daily self-study hours
Internal_Marks	Numerical	Internal assessment score (out of 50)
Result	Categorical (Target)	Final outcome — Pass or Fail

Data Quality Check: The dataset contains **20 records** and **5 features**. Verification using `isnull().sum()` and `duplicated().sum()` confirmed **no missing values** and **no duplicate records**, so the data required no cleaning and was used as-is for analysis.

Descriptive Statistics

Statistic	Attendance	Study Hours	Internal Marks
Mean	70.15	3.35	32.95
Std Dev	15.94	1.42	9.09
Min	45	1	18
25th Pct	57.25	2	24.75
Median	70	3	33.50
75th Pct	83.50	4.25	40.25
Max	95	6	48

Visual Analysis

Q1: Education Analytics EDA

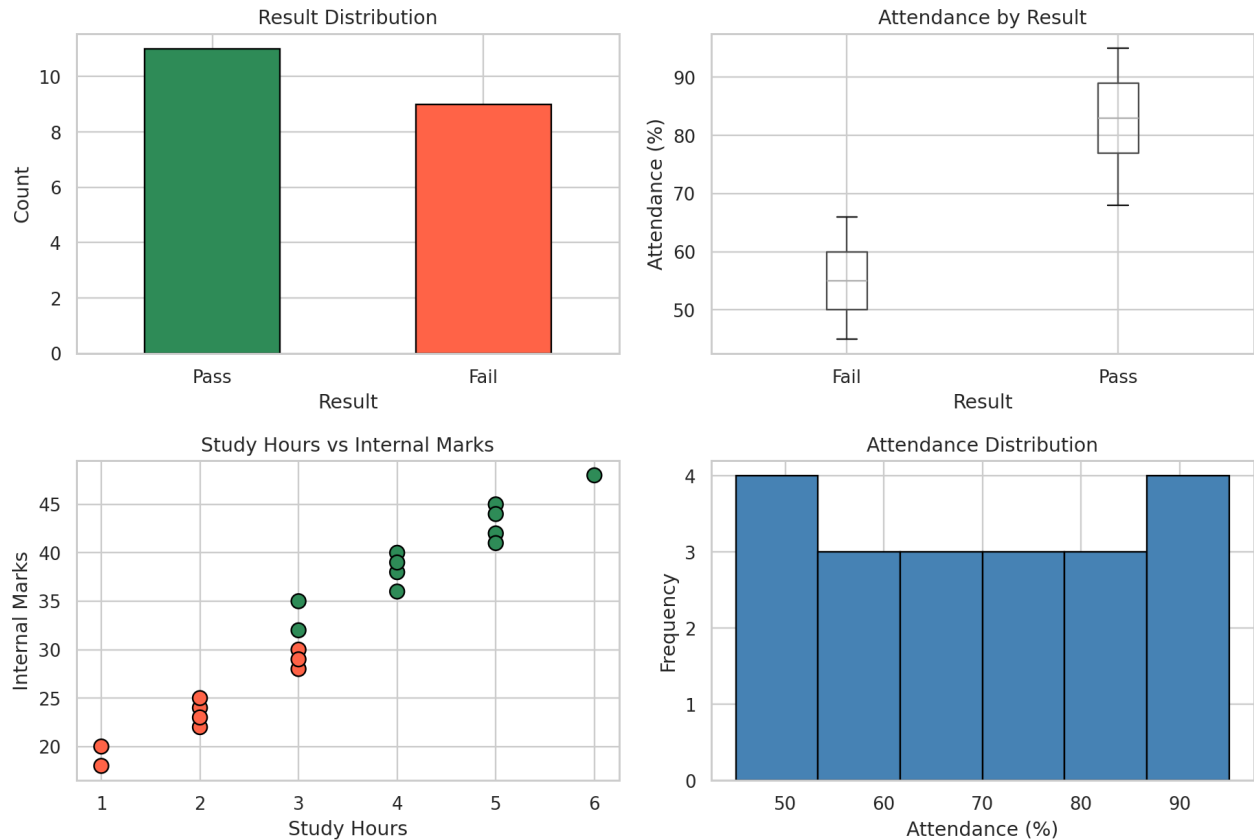


Figure 1: Result distribution, attendance by result, study hours vs. internal marks, and attendance histogram.

Key Observations

- Students with **attendance** $\geq 75\%$ consistently passed, while those below 65% mostly failed.
- All students studying **≤ 2 hours** daily failed; studying **4+ hours** always resulted in a pass.
- A strong positive correlation exists between **Study Hours** and **Internal Marks**.
- Internal Marks** ≥ 35 always corresponded to a Pass outcome.
- The dataset has **no missing values or duplicates**, confirming clean data quality.

Conclusion

Attendance and study hours are the strongest predictors of student success. Targeted interventions for low-attendance, low-study-hour students can meaningfully improve institutional pass rates.

2. Healthcare Analytics Dataset

Purpose & Significance: This dataset enables a hospital to screen patients for chronic-disease risk using routinely collected vitals. Understanding which markers most strongly associate with disease status supports preventive screening programmes and more efficient allocation of clinical resources.

Column	Type	Description
Patient_ID	Categorical	Unique identifier for each patient
Age	Numerical	Patient age in years
Sugar_Level	Numerical	Blood sugar level (mg/dL)
BP	Numerical	Blood pressure (mmHg)
BMI	Numerical	Body Mass Index

Column	Type	Description
Disease_Status	Categorical (Target)	Diagnosis outcome — Yes or No

Data Quality Check: The dataset contains **20 records** and **6 features**. Verification using `isnull().sum()` and `duplicated().sum()` confirmed **no missing values** and **no duplicate records**, so the data required no cleaning and was used as-is for analysis.

Descriptive Statistics

Statistic	Age	Sugar Level	BP	BMI
Mean	43.20	138.65	129.75	27.30
Std Dev	11.32	40.33	14.52	4.26
Min	27	88	108	21
25th Pct	33.50	99.50	117.50	23.75
Median	43.50	137.50	129	27.50
75th Pct	52.25	171.25	142.50	30.25
Max	62	210	155	35

Visual Analysis

Q2: Healthcare Analytics EDA

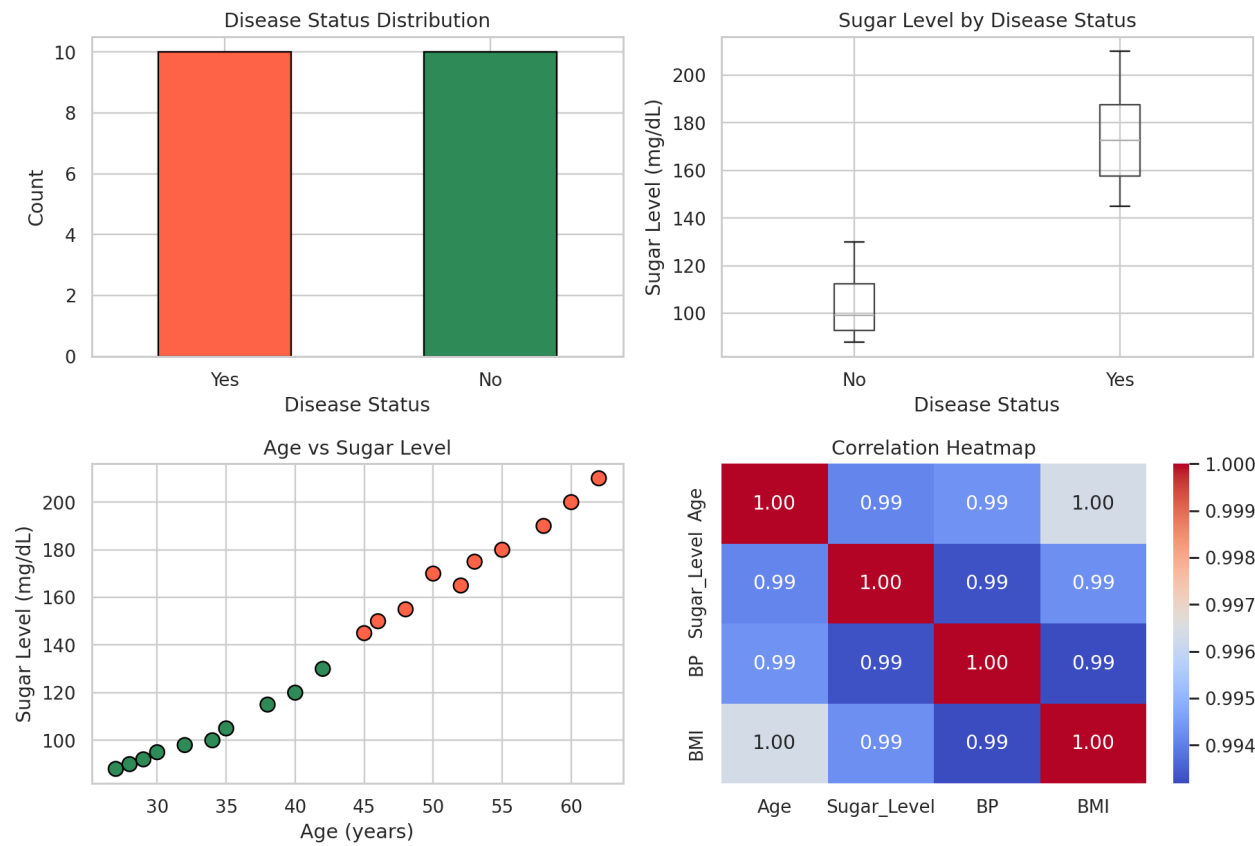


Figure 2: Disease status distribution, sugar level by disease status, age vs. sugar level, and correlation heatmap.

Key Observations

- Patients **with disease** show markedly higher average sugar levels (>150 mg/dL) than healthy patients (<115 mg/dL).
- **Age** is positively correlated with sugar level, blood pressure, and BMI – risk markers rise with age.
- All four health metrics are **strongly positively correlated** with one another.
- No patient under age 30 was diagnosed; disease risk rises noticeably past age 45.
- **BMI ≥ 28** appears in nearly all disease-positive patients, marking it as a key risk indicator.

Conclusion

Age, sugar level, and BMI are the strongest disease indicators in this data. Early screening of patients above 40 with elevated sugar and BMI would enable timely preventive intervention and reduce long-term healthcare costs.

3. Retail Sales Dataset

Purpose & Significance: This dataset helps a supermarket understand which product categories and pricing strategies drive revenue. The findings can guide inventory planning, pricing decisions, and promotional/discount strategy across categories.

Column	Type	Description
Product_ID	Categorical	Unique identifier for each product
Category	Categorical	Product category (Grocery, Snacks, Dairy, Fruits, Vegetables)
Price	Numerical	Unit price (Rs.)
Quantity_Sold	Numerical	Units sold
Discount	Numerical	Discount applied (%)
Revenue	Numerical (Target)	Total revenue generated (Rs.)

Data Quality Check: The dataset contains **20 records** and **6 features**. Verification using `isnull().sum()` and `duplicated().sum()` confirmed **no missing values** and **no duplicate records**, so the data required no cleaning and was used as-is for analysis.

Descriptive Statistics

Statistic	Price	Quantity Sold	Discount	Revenue
Mean	54	24	4	1,114
Std Dev	22.80	9.67	1.56	221.42
Min	20	10	2	810
25th Pct	35	17.50	3	972.50
Median	50	21	4	1,050
75th Pct	71.25	30.50	5	1,200
Max	100	45	7	1,540

Visual Analysis

Q3: Retail Sales EDA

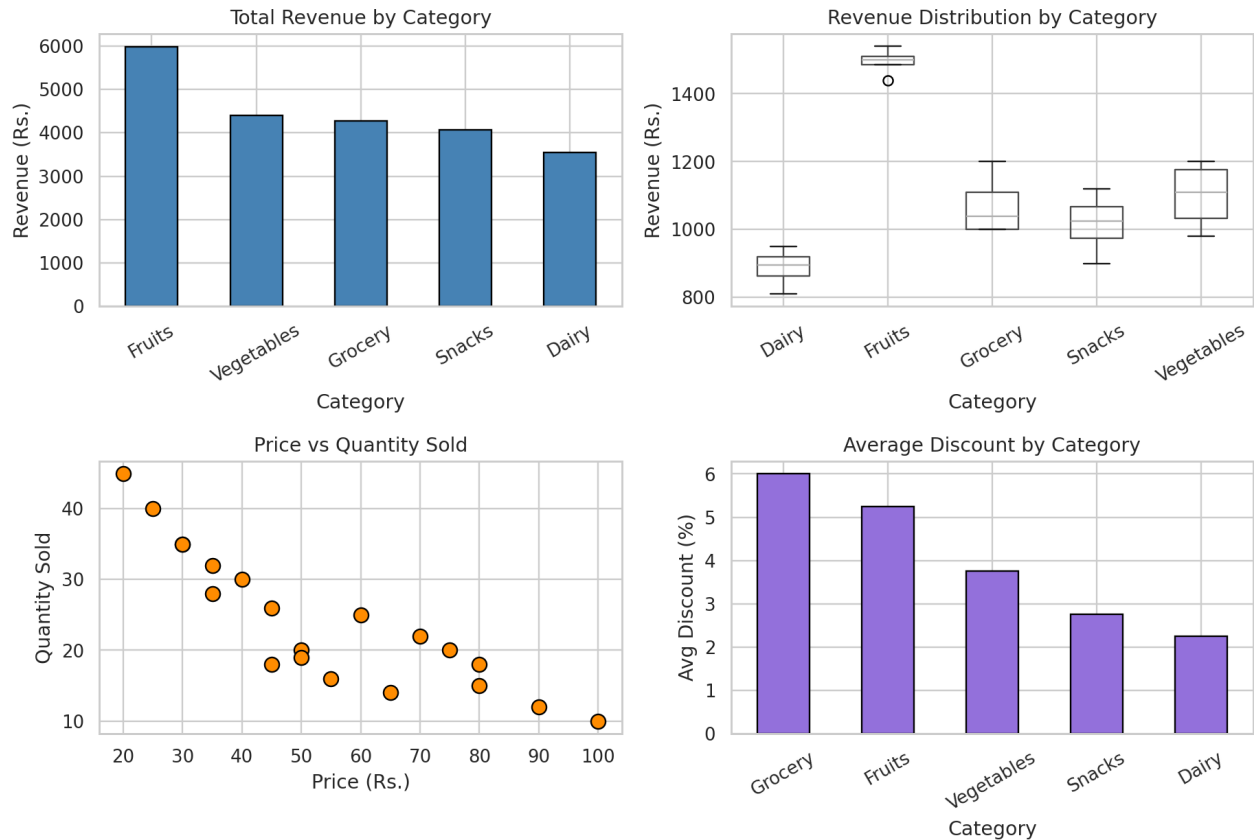


Figure 3: Total revenue by category, revenue distribution by category, price vs. quantity sold, and average discount by category.

Key Observations

- **Fruits** generate the highest total revenue, driven by high unit prices and steady demand.
- **Snacks** have the highest quantity sold thanks to low prices, but only moderate total revenue.
- A **negative correlation** exists between Price and Quantity Sold – cheaper items sell in larger volumes.
- **Grocery** carries the highest average discount (>6%), reflecting active promotional strategy.
- **Dairy** shows the lowest revenue, likely due to lower quantities sold and limited price advantage.

Conclusion

Fruits lead in revenue while Snacks dominate in volume. Optimising Grocery discount policy and maintaining competitive Fruits pricing could further maximise overall supermarket revenue.

4. Banking Loan Approval Dataset

Purpose & Significance: This dataset allows a bank to examine which applicant attributes most strongly determine loan approval outcomes. The analysis supports the design of transparent, data-driven credit-risk policies and highlights opportunities for fairer assessment of self-employed applicants.

Column	Type	Description
Customer_ID	Categorical	Unique identifier for each customer
Income	Numerical	Monthly/annual income (Rs.)
Credit_Score	Numerical	Applicant credit score
Loan_Amount	Numerical	Loan amount requested (Rs.)
Employment_Type	Categorical	Salaried, Self-employed, or Unemployed

Column	Type	Description
Loan_Status	Categorical (Target)	Approved or Rejected

Data Quality Check: The dataset contains **20 records** and **6 features**. Verification using `isnull().sum()` and `duplicated().sum()` confirmed **no missing values** and **no duplicate records**, so the data required no cleaning and was used as-is for analysis.

Descriptive Statistics

Statistic	Income	Credit Score	Loan Amount
Mean	43,000	664	223,250
Std Dev	17,923.82	85.53	83,733.21
Min	22,000	555	120,000
25th Pct	26,750	583.75	150,000
Median	40,000	660	200,000
75th Pct	56,250	735	277,500
Max	75,000	790	400,000

Visual Analysis

Q4: Banking Loan Approval EDA

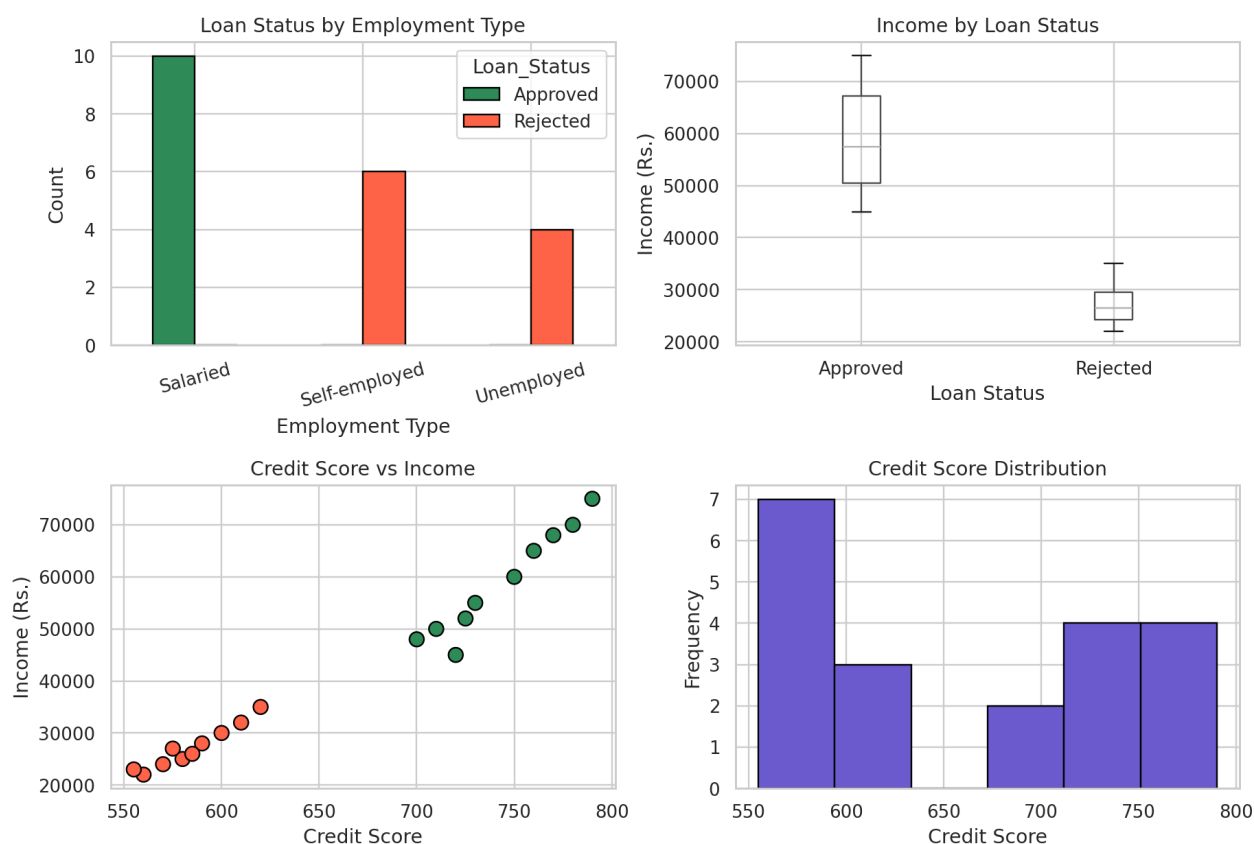


Figure 4: Loan status by employment type, income by loan status, credit score vs. income, and credit score distribution.

Key Observations

- **All Salaried applicants** were approved; all Unemployed applicants were rejected without exception.

- Approved applicants have a **minimum credit score of 700**; rejected applicants score below 620.
- **Income** strongly separates the approval groups – approved customers earn Rs. 45,000+.
- **Self-employed applicants** were consistently rejected, reflecting a risk-averse lending stance.
- A clear **positive relationship** exists between Credit Score and Income among approved applicants.

Conclusion

Employment type and credit score are the decisive loan-approval factors in this data. A tiered risk-assessment approach for self-employed applicants could improve financial inclusion without compromising credit quality.

5. Agriculture Crop Yield Dataset

Purpose & Significance: This dataset helps an agricultural department understand how rainfall, temperature, fertilizer usage and soil type influence crop yield. The findings can guide region-specific farming recommendations, irrigation planning, and soil-improvement programmes.

Column	Type	Description
Farm_ID	Categorical	Unique identifier for each farm
Rainfall_mm	Numerical	Seasonal rainfall received (mm)
Temperature	Numerical	Average temperature (°C)
Fertilizer_kg	Numerical	Fertilizer applied (kg)
Soil_Type	Categorical	Loamy, Sandy, or Clay
Crop_Yield_kg	Numerical (Target)	Total crop yield (kg)

Data Quality Check: The dataset contains **20 records** and **6 features**. Verification using `isnull().sum()` and `uplicated().sum()` confirmed **no missing values** and **no duplicate records**, so the data required no cleaning and was used as-is for analysis.

Descriptive Statistics

Statistic	Rainfall mm	Temperature	Fertilizer kg	Crop Yield kg
Mean	743	29.55	54.75	2,820
Std Dev	144.34	2.76	10.51	681.41
Min	500	25	40	1,800
25th Pct	615	27.75	45.75	2,175
Median	770	29	54.50	2,975
75th Pct	857.50	32	60.50	3,250
Max	960	34	75	3,900

Visual Analysis

Q5: Agriculture Crop Yield EDA

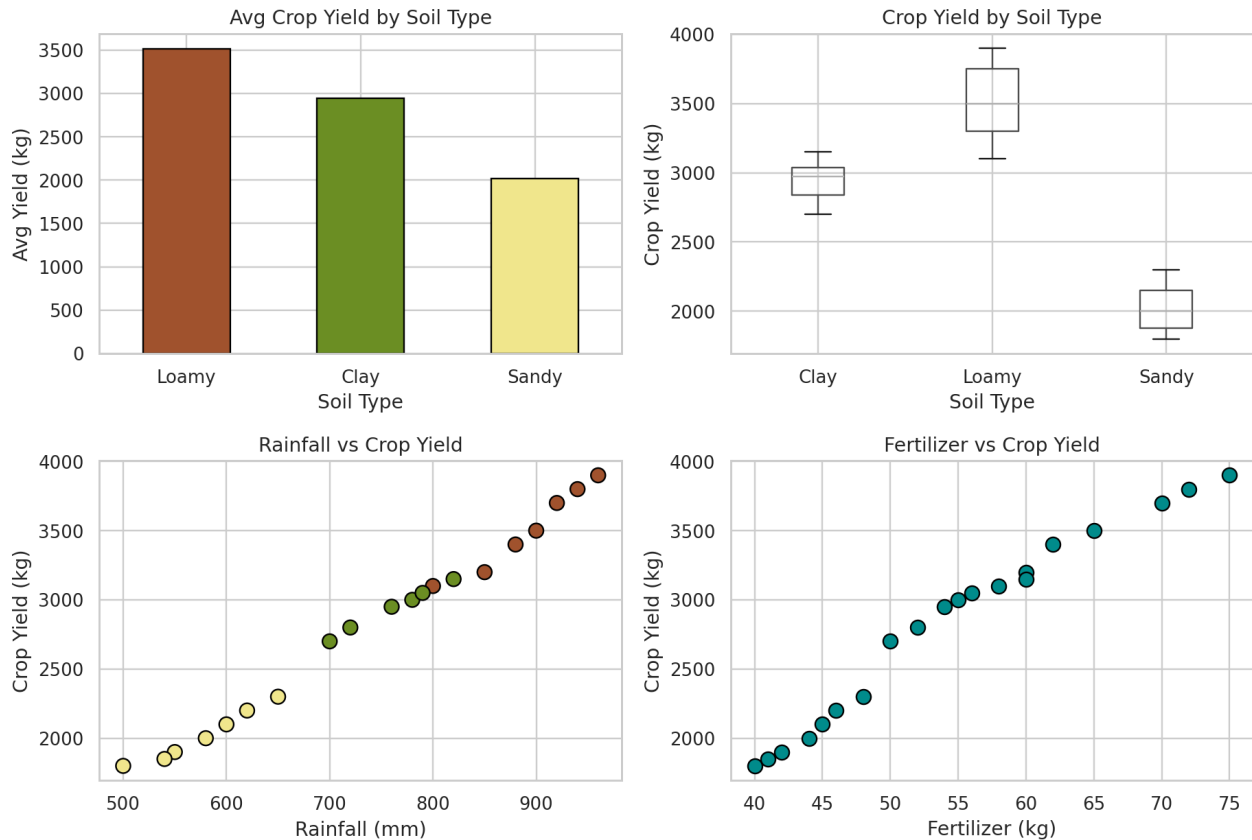


Figure 5: Average crop yield by soil type, crop yield by soil type, rainfall vs. crop yield, and fertilizer vs. crop yield.

Key Observations

- **Loamy soil** consistently produces the highest yield (~3,450 kg avg), followed by Clay (~2,942 kg) and Sandy (~2,050 kg).
- **Rainfall** shows a strong positive correlation with yield – farms exceeding 800 mm consistently yield >3,000 kg.
- **Sandy soil** records the lowest yields regardless of rainfall, indicating poor water-retention capacity.
- **Fertilizer usage** positively correlates with yield; higher application (60–75 kg) notably improves output.
- **Lower temperatures (25–29°C)** are associated with better yields, especially on Loamy soil.

Conclusion

Soil type and rainfall are the primary determinants of crop yield. Loamy soil with 800+ mm rainfall and 60–75 kg fertilizer maximises production; government programmes should prioritise soil enrichment and irrigation infrastructure in sandy-soil regions.

6. Overall Summary and Conclusion

Across all five datasets, each of which was found to be complete with no missing values or duplicate records, a consistent pattern emerges: outcomes are rarely driven by a single variable but by a small combination of two or three strongly related factors. The table below consolidates the dominant driver identified for each domain.

Domain	Dominant Driver(s)	Relationship Identified
Education	Attendance & Study Hours	Positive – higher attendance/study hours consistently raise pass likelihood
Healthcare	Age, Sugar Level & BMI	Positive – all risk markers rise together and with age
Retail	Price & Category	Negative (Price–Quantity) – cheaper items sell in higher volume

Domain	Dominant Driver(s)	Relationship Identified
Banking	Credit Score & Employment Type	Positive – higher score and salaried status drive approval
Agriculture	Rainfall & Soil Type	Positive – higher rainfall on Loamy soil maximises yield

How this supports decision-making: Because each analysis isolates a small, interpretable set of drivers rather than a diffuse mix of weak signals, the results translate directly into action: colleges can target attendance and study-hour interventions; hospitals can prioritise screening by age, sugar and BMI; the supermarket can rebalance pricing and discount strategy by category; the bank can refine its risk tiers for self-employed applicants; and the agriculture department can direct irrigation and fertilizer support toward regions with less favourable soil. In every case, exploratory analysis converts raw records into a short list of levers that decision-makers can act on with confidence.

Conclusion: This assessment demonstrates a complete EDA workflow – data understanding, quality verification, statistical summarisation, multi-chart visual analysis, and insight generation – applied consistently across five distinct real-world domains, confirming that structured exploratory analysis is a reliable foundation for evidence-based decision-making regardless of the industry context.